

RRC VN MACHINE SAFETY & VALIDATION: SEMI-AUTOMATIC RESISTANCE SPOT WELDING MACHINE

Safety, Process & Product Qualification

Abstract

This document validates the safety and functionality of the modified resistance spot welding machine with its automated X-Y positioning system. It covers machine safety, process control, welding accuracy, traceability, fault recovery, and final welding quality to demonstrate that the machine meets the defined requirements and is suitable for its intended production use.

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1 Purpose

The purpose of this document is to validate the safety and functionality of the modified semi-automatic resistance spot welding machine before it is released for production. The document also provides evidence of compliance with applicable Vietnamese occupational safety requirements and documents the application of selected European machine safety standards as engineering guidance.

2 Machine Description

The machine is a **semi-automatic resistance spot welding machine** created by integrating an X-Y positioning table into an existing manual resistance spot welding machine. The original welding unit remains unchanged. The modification automates the positioning of the workpiece while the original welding process is retained.

During operation, the operator loads the workpiece into the welding fixture and presses the **START** button. The X-Y positioning table automatically moves the workpiece through all programmed welding positions. After the welding cycle is completed, the table returns to the load/unload position, where the finished workpiece is removed and the next workpiece is loaded.

The machine is equipped with appropriate safety devices to protect the operator during automatic operation. The safety concept and validation are described in this document.

Machine operation is controlled by a Python application running on “Station 3 NUC industrial PC”. The software communicates with the X-Y motion controller via an RS232 interface and with the welding machine through an Ethernet-connected digital I/O module. The digital I/O interface is used to:

- Select the required welding program,
- Start the welding cycle,
- Receive machine status and welding result signals, and
- Monitor machine operating conditions required for the automatic process.



Photo of the Semi-Automatic Resistance Spot Welding Machine

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The digital input and output signals used for machine control are shown in **Figure 1 - Machine Control Architecture** and are verified during the functional validation described in this document.

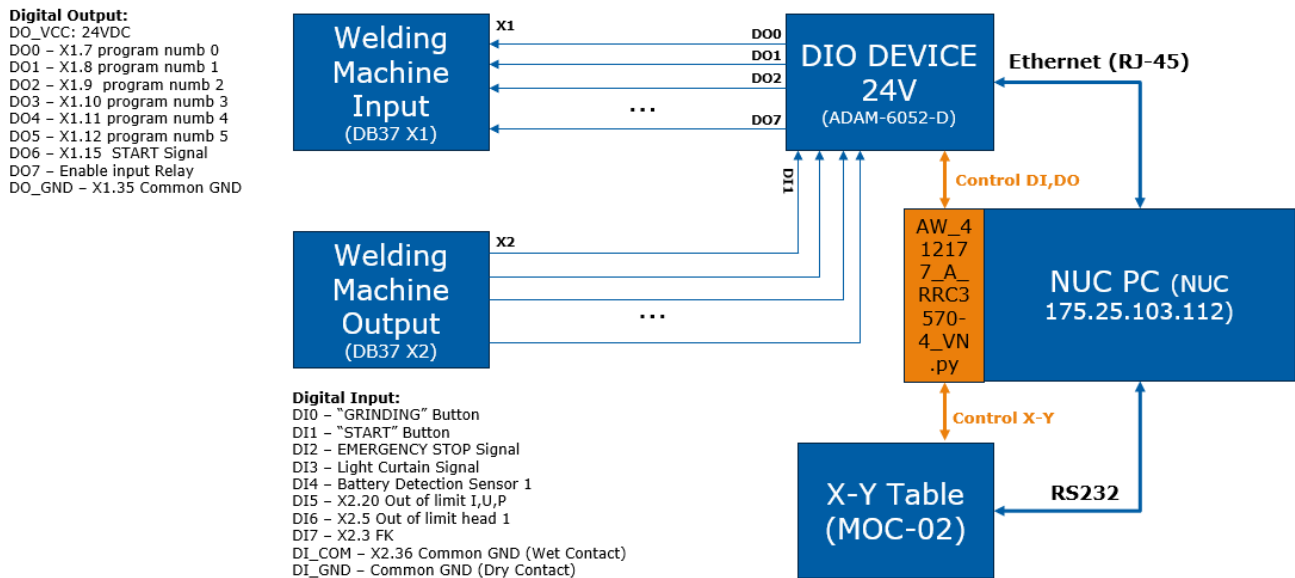


Figure 1 - Machine Control Architecture

Machine Main Components

Component	Manufacturer	Model	Remarks
Weld Head	AMADA WELD TECH GmbH, Lindberghstr. 1, 82178 Puchheim, Germany	F120/A – 145	Manuf.No. 400507877- 2/08/22
Inverter Power Source	AMADA WELD TECH GmbH, Lindberghstr. 1, 82178 Puchheim, Germany	IS-Q250A- EXT-AC/DC (Rev.05) 22 KVA	Manuf.No. SH013173- 05S0974
Resistance spot welding machine controller	AMADA WELD TECH GmbH, Lindberghstr. 1, 82178 Puchheim, Germany	AWS3- Control Panel / AMADA MIYACHI OP-AWS3-A	Manuf. No. A11081D- ADVANCED
X-Y positioning table / Motorized XY-Axis Stage	Optics Focus Instruments Co., Ltd., Room 601, JinTaiLiDa Building, NO. 4 South Road, JiuXianQiao, Chaoyang District 100016, Beijing	MOXY-06- 300-300	Added during modification
X-Y controller / 2- Axis Motion Controller	Optics Focus Instruments Co., Ltd., Room 601, JinTaiLiDa Building, NO. 4 South Road, JiuXianQiao, Chaoyang District 100016, Beijing	MOC-02-2- 220	Controls table movement
NUC PC	Wortmann AG, Bredenhop 20, 32609 Hüllhorst, Germany	TERRA PC- Micro 6000 SILENT GREENLINE	NUC PC of station 3 is used for this machine

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DIO / Remote Ethernet I/O Module	Advantech Vietnam Technology Co., Ltd., 2nd Floor, VTC Online Building, 18 Tam Trinh Street Tuong Mai Ward Hanoi, Vietnam	ADAM-6052-D	I/O hardware between the PC control software and the machine
Safety relay	Pilz	PN0Z X2.8P	Safety relay module
Safety contactors	Schneider Electric	LC1D09M7	Contactor, TeSys Deca, 3P(3NO), AC-3/3e, <=440V, 9A, 220V AC 50/60Hz coil, screw clamp terminals
Safety light curtain	Pilz	PSEN opII4F-s-14-060	Safety light guard, Type 4, PL e, finger protection (resolution 14 mm), protected field height 600 mm without dead zone, supply voltage 24 V. Connector M12, transmitter: 5-pin, receiver 5-pin.
Emergency stop	IDECC	YW1B-V4E02R	Mushroom pushbutton

2.1 Vietnam Legal Compliance

Vietnam does not have a general Machinery Directive comparable to the EU Machinery Directive. Machine safety is primarily governed by occupational safety requirements. Compliance must be demonstrated by the employer and operator. Certain machine categories are additionally subject to technical inspection and applicable QCVN technical regulations.

2.1.1 Vietnam Legal Requirements & Vietnam Compliance Checklist

Requirement	Reference	Status	Evidence / Comments
Status symbols used like this in the entire document: ☒ = Verified / Compliant ☐ = Not verified / Open N/A = Not applicable			
Legal requirements identified	Law No. 84/2015/QH13 = Occupational Safety and Hygiene Law	☒	E1
	Decree No. 44/2016/ND-CP = Implementation Decree	☒	E2
	Circular No. 54/2016/TT-BLDTBXH = Technical inspection procedures	☒	E3

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Risk assessment completed	ISO 12100 – technical reference only	☒	E4
Safe operating procedure available	Company Procedure	☒	E5
Operator training completed	Company Procedure / ASI	☒	E6
Emergency stop verified	ISO 13850 – technical reference only	☒	E15
Light curtain installed and tested	IEC 61496 – technical reference only	☒	E7
Electrical Safety Verification	IEC 60204-1 – technical reference only	☒	E8
Pneumatic hazards assessed	ISO 4414 – technical reference only	☒	E9
Maintenance plan defined	Company Procedure	☒	0

2.2 CE Compliance

This section is provided as best practice guidance for the design and validation of internally developed machines.

The machine is designed, built and operated in Vietnam and is therefore not legally required to comply with the European Machinery Directive 2006/42/EC or the Machinery Regulation (EU) 2023/1230. Compliance with these European regulations is not claimed by this document.

RRC Power Solutions Co., Ltd. uses selected European machinery safety principles and harmonized standards as engineering guidance to achieve a high level of machine safety. **The checklists in this section are intended to support the design review and validation process and should not be interpreted as a declaration of full CE compliance or certification.**

Where applicable, individual requirements may be adopted in full, adopted in part, or documented as not applicable based on the machine design, risk assessment, intended use, and local legal requirements.

Requirement	Reference	Status	Evidence / Comments
Risk assessment completed	ISO 12100	☒	E11
Safety-Related Control System reviewed against ISO 13849-1 principles	ISO 13849-1	☒	E12
Safety functions validated	ISO 13849-2	☒	E13 / E14 / E15 / E16
Electrical equipment reviewed	IEC 60204-1	☒	E8
Emergency stop validated	ISO 13850	☒	E15
Safety Distance Verification	ISO 13855	☒	E4

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Interlocking devices reviewed (if applicable)	ISO 14119	N/A	No interlocking devices installed.
Machine documentation completed	Company Procedure	☒	E4/ E5 / 0
Operating instructions available	Company Procedure	☒	E5

2.3 Machine Safety Validation Checklist

2.3.1 Mechanical Safety

Check	Status	Evidence / Comments
Welding electrode hazards assessed	N/A	Original CE-certified welding unit (welding head, electrodes and pneumatic Z-axis) not modified; outside the scope of this validation.
Pneumatic Z-axis assessed	N/A	E9
Emergency stops accessible	☒	E15
Safety distances verified	☒	E16

2.3.2 Electrical Safety

Check	Status	Evidence / Comments
Grounding verified	☒	E8
Protection devices verified	☒	E8
Power isolation verified	☒	E8
Electrical cabinet inspected	☒	E8

2.3.3 Fault Simulation

Check	Status	Evidence / Comments
Broken light curtain by cable fault = broken light curtain & cable fault	☒	E13
Power loss / restore & Unexpected restart	☒	E14 / E7 / E15

2.4 Machine Functional Validation

The purpose of the functional validation is to verify that the modified semi-automatic resistance spot welding machine consistently performs the intended manufacturing process and continues to produce products that meet the specified quality requirements.

The validation confirms that the modification from a manual to a semi-automatic welding process has been implemented in a controlled manner and that the machine continues to produce conforming products **under normal operating conditions as well as during defined abnormal or fault conditions**.

The functional validation focuses on process-related functions that may influence product quality, process reliability, traceability, or the detection of non-conforming products. This includes verification

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of the welding process, process control, error handling, controlled recovery after interruptions, and the recording of production and quality data.

The detailed validation activities and their results are documented in the corresponding evidence sections of this report.

2.4.1 Functional Validation Overview

Functional Validation Item	Focus	Evidence
Workpiece Preparation & Fixture Validation	Correct workpiece loading, cell-tab positioning, clamping and fixture repeatability	E17
Welding Position & Process Parameter Verification	Correct weld position and correct welding program selection	E18
Welding Result & NOK Handling	Out-of-limit detection, process stop, machine lock, supervisor release	E19
Process Interruption & Controlled Recovery	Emergency stop, safety light curtain, power loss, controlled recovery, no skipped welds and no duplicate welds after restart	E20
Product Traceability & Welding Data	Work Order, UDI and welding data recording (OK and NOK)	E21
Final Product Validation	Five complete battery packs, complete welding sequence (42 welds), exactly-once welding , visual inspection, welding monitor evaluation, pull testing and metallographic section analysis	E23

2.5 Document Approval

2.5.1 VALIDATION EVIDENCE – RESPONSIBILITY & SIGN-OFF

By signing, the responsible person confirms that the respective validation evidence (“E”) has been completed, reviewed and is acceptable. Signatures apply to the evidence section identified by the E-number (shown as “E_” in the table header below) and its complete associated evidence package.

E_	Validation Evidence	Responsible Person	Signature	Date
E1		Hang Nguyen		28 th Aug.2026

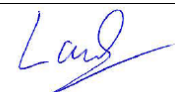
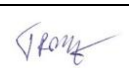
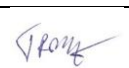
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	Law No. 84/2015/QH13 – Law on Occupational Safety and Hygiene	Ingo Bauerschmidt		28 th Aug.2026
E2	Decree No. 44/2016/ND-CP – Implementation Decree	Hang Nguyen		28 th Aug.2026
		Ingo Bauerschmidt		28 th Aug.2026
E3	Circular No. 54/2016/TT-BLDTBXH – Technical inspection procedures	Hang Nguyen		28 th Aug.2026
		Ingo Bauerschmidt		28 th Aug.2026
E4	Machine Risk Assessment	Huong Pham		28 th Aug.2026
		Khanh Le		28 th Aug.2026
E5	Safe Operating Procedure	Anh Nguyen		28 th Aug.2026
E6	Operator Training	Hang Nguyen		28 th Aug.2026
		Hung Dang		28 th Aug.2026
E7	Safety Light Curtain	Cuong Nguyen		28 th Aug.2026
		Duc Bui		28 th Aug.2026
E8	Electrical Safety	Cuong Nguyen		28 th Aug.2026
E9	Pneumatic Safety	Cuong Nguyen		28 th Aug.2026
		Hung Dang		28 th Aug.2026
E10	Maintenance Plan	Cuong Nguyen		28 th Aug.2026
E11	Risk Assessment (ISO 12100)	Hung Dang		28 th Aug.2026

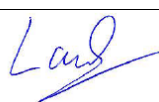
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E12	Safety-Related Control System Review (ISO 13849-1)	Cuong Nguyen		28 th Aug.2026
		Hung Dang		28 th Aug.2026
E13	Light Curtain Fault Simulation	Cuong Nguyen		28 th Aug.2026
		Hung Dang		28 th Aug.2026
E14	Power Loss & Restore	Cuong Nguyen		28 th Aug.2026
E15	Emergency Stop	Cuong Nguyen		28 th Aug.2026
E16	Safety Distance Verification	Cuong Nguyen		28 th Aug.2026
		Lam Vo		28 th Aug.2026
E17	Workpiece Preparation & Fixture Validation	Duc Bui		28 th Aug.2026
		Khanh Le		28 th Aug.2026
		Trong Dang		28 th Aug.2026
E18	Welding Position & Process Parameter Verification	Anh Nguyen		28 th Aug.2026
		Khanh Le		28 th Aug.2026
		Trong Dang		28 th Aug.2026
E19	Welding Result & NOK Handling	Anh Nguyen		28 th Aug.2026
		Lam Vo		28 th Aug.2026

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E20	Process Interruption & Controlled Recovery	Lam Vo		28 th Aug.2026
		Anh Nguyen		28 th Aug.2026
E21	Product Traceability & Welding Data Recording	Lam Vo		28 th Aug.2026
E22	N/A	N/A	N/A	N/A
E23	Final Product Validation	Khanh Le		28 th Aug.2026
		Trong Dang		28 th Aug.2026

2.5.2 DOCUMENT APPROVAL

By signing, the persons listed below confirm that this validation document has been reviewed and approved. The validation results and associated evidence have been reviewed and the machine is accepted for its intended use, subject to all specified acceptance criteria being fulfilled.

	Name	Signature
Prepared by	Hung Dang & Ingo Bauerschmidt	  28 th Aug.2026
Approved by	Ingo Bauerschmidt	 28 th Aug.2026

2.6 Support documents and validation evidence (“E1 ... En”)

E1 - Law No. 84/2015/QH13 – Law on Occupational Safety and Hygiene

Law No. 84/2015/QH13 – Law on Occupational Safety and Hygiene was checked on 07.08.2026 by RRC VN HR & EHS specialist and does not specify detailed machine safety requirements comparable to the European Machinery Directive or Machinery Regulation. Instead, it establishes the general obligation of the employer to ensure that machinery is safe and that employees are adequately protected.

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E2 - Decree No. 44/2016/ND-CP = Implementation Decree

The corresponding implementing decrees and circulars define additional requirements, such as operator training, technical inspection, and machinery subject to mandatory occupational safety inspection. Compliance with these general legal obligations is demonstrated by the evidence in E4–E11.

E3- Circular No. 54/2016/TT-BLĐTBXH = Technical inspection procedures

Circular No. 54/2016/TT-BLĐTBXH was reviewed to determine whether the modified semi-automatic resistance spot welding machine is subject to mandatory technical safety inspection in accordance with Vietnamese occupational safety legislation.

The machine was reviewed against the categories of machinery and equipment requiring mandatory technical safety inspection. At the time of this assessment, the modified semi-automatic resistance spot welding machine was **not identified as equipment subject to mandatory technical safety inspection** under the applicable Vietnamese regulations.

Therefore, the technical inspection procedures defined by Circular No. 54/2016/TT-BLĐTBXH are **not applicable** to this machine.

E4 – Machine Risk Assessment

Hazard Category	Hazards Considered	Protective Measures Implemented
Mechanical Hazards	Moving X-Y table, pinch and crush points, welding electrode movement (Z-axis)	Safety light curtain, emergency stop, safety relay, safety contactors, machine guarding, safe operating procedure
Electrical Hazards	220 VAC power supply, electrical cabinet, grounding and electrical protection	Protective earth (PE), electrical protection devices, enclosed electrical cabinet, controlled access
Pneumatic Hazards	Unexpected movement due to compressed air, loss or restoration of air pressure	Pneumatic system designed for safe operation, controlled restart procedure
Thermal Hazards	Hot welding electrodes	Operator awareness, safe operating procedure
Operational Hazards	Incorrect loading or unloading of the workpiece, access to the hazardous area during automatic operation	Light curtain, emergency stop, manual reset, operator training
Safety Devices	Light curtain, emergency stop, Pilz PNOZ X2 safety relay, safety contactors	Functional validation and periodic verification (Maintenance & EHS)
Residual Risks	Remaining risks during operation and maintenance	Operating instructions, operator training
Overall Assessment	☒ Risks identified, evaluated and reduced to an acceptable level	

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A machine risk assessment has been performed for the semi-automatic resistance spot welding machine. The identified mechanical, electrical, pneumatic, thermal and operational hazards have been evaluated and appropriate protective measures have been implemented.

The safety concept includes a safety light curtain, emergency stop, Pilz PNOZ X2 safety relay, safety contactors, and a manual restart sequence. Residual risks have been reduced to an acceptable level by technical protective measures, operating instructions, operator training, and the use of personal protective equipment (PPE) where required.

The results of this assessment support compliance with the general machine safety obligations defined by **Law No. 84/2015/QH13 – Law on Occupational Safety and Hygiene**. The assessment will be reviewed whenever significant modifications are made to the machine or its intended use.

E5 - Safe Operating Procedure

Objective: To verify that a documented safe operating procedure is available for the semi-automatic resistance spot welding machine.

The safe operation of the machine is covered by the approved **Assembly Instruction (ASI)** for the workstation A03.

The ASI includes the following topics:

No.	Topic	Status
1	Machine description and intended use	☒
2	Operator qualifications	☒
3	Required personal protective equipment (PPE)	☒
4	Pre-operation inspection	☒
5	Workpiece loading procedure	☒
6	Starting the welding cycle	☒
7	Normal operating sequence	☒
8	Workpiece unloading procedure	☒
9	Emergency stop procedure	☒
10	Safety light curtain operation	☒
11	Restart procedure after a safety stop	☒
12	Machine shutdown procedure	☒
13	Cleaning instructions	☒
14	Prohibited actions and safety precautions	☒

Result: The Assembly Instruction (ASI) for the workstation has been reviewed and verified to ensure that all safety-related operating instructions required for the safe operation of the semi-automatic resistance spot welding machine are included.

The ASI is maintained as a controlled document within the RRC Quality Management System. Any revision of the ASI shall be reviewed to ensure that the documented safety instructions remain valid for the current machine configuration.

Referenced document:

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AA_F_022_VN_B_Resistance-welder_Released

E6 – Operator Training

Objective: To verify that all authorized operators have received the required training before operating the semi-automatic resistance spot welding machine.

The required training is based on the approved **Assembly Instruction (ASI)** for the respective workstations.

The training includes all safety-related and operational topics defined in the ASI, including:

- Machine operation
- Safety precautions
- Emergency stop procedure
- Safety light curtain operation
- Restart procedure after a safety stop
- Required personal protective equipment (PPE)
- All additional instructions defined in the current ASI revision

Result: Training has been completed for all authorized operators before operating the machine. The training records are maintained as controlled quality records within the RRC Quality Management System.

Referenced documents: Training ASI RRC3570-4 Pre-assembly Rev.H: Record No. [1794](#)

E7 - Safety Light Curtain

Objective: To verify that the safety light curtain is correctly installed and performs its intended safety function.

Installation Verification

Check	Status
Certified safety light curtain installed	☒
14 mm detection capability verified	☒
Dual-channel OSSD outputs connected correctly	☒
Connected to Pilz PNOZ X2 safety relay	☒
Protective field correctly positioned	☒
Safety distance verified (see 0)	☒
Hazardous area completely protected	☒

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Photo of safety light curtain installation.



Manufacturer's data of the light curtain

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Functional Test: the following test was performed **5 times**.

Test	Expected Result	Result
Interrupt the light curtain during X-Y table movement	PNOZ X2 switches to safe state	☒ PASS
	The safety circuit removes the power supply to the X-Y motor driver stages. The X-Y motion controller remains powered for communication and control functions, but hazardous X-Y movement is disabled.	☒ PASS
	X-Y table stops	☒ PASS
	The computer monitor shows a notification and prompts for confirmation	☒ PASS
Press START	Machine operation can only resume after completion of the defined recovery procedure and a manual START command.	☒ PASS

Acceptance Criteria

- All 5 tests shall pass successfully.
- Every interruption of the protective field shall activate the safety function.
- No automatic restart shall be possible.

Assessment Result: The safety light curtain has been verified for correct installation and functionality. The protective field reliably activates the safety function whenever it is interrupted. The protective field reliably activates the safety function whenever it is interrupted. The machine remains in a safe state until the hazardous area is clear, a notification message appears on the screen requiring operator confirmation to restore the machine to the ready state, and then the Machine operation can only resume after completion of the defined recovery procedure and a manual START command.

E8 - Electrical Safety

Objective: To verify that the electrical modifications associated with the X-Y positioning table and the safety system have been installed correctly and operate safely.

Installation Verification

Check	Status
Electrical installation inspected	☒
Protective earth (PE) and power supply verified	☒
Safety circuit correctly wired and tested	☒
Wiring and cable identification verified	☒

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Photos of the electrical cabinet

Phase L1-N-PE



Phase L2-N-PE



Phase L3-N-PE



Photo of PE measuring

Assessment Result: The electrical modifications associated with the X-Y positioning table and the safety system have been inspected and functionally verified. The electrical installation is complete and correctly wired.

The original electrical installation of the CE-certified resistance spot welding machine has not been modified and is therefore outside the scope of this validation.

E9 – Pneumatic Safety

Objective: To verify whether modifications to the pneumatic system require validation as part of this project.

Assessment: The pneumatic system of the original resistance spot welding machine, including the pneumatic Z-axis, valves, cylinders, and associated pneumatic safety functions, has **not been modified** as part of this project.

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The scope of this project is limited to the addition of the X-Y positioning table and the related electrical safety system, including the safety light curtain, emergency stop, safety relay, and safety contactors.

Therefore, no changes have been made to the existing pneumatic system and no additional pneumatic safety validation is required.

Evidence: Comparison of the original machine configuration with the modified machine confirms that no pneumatic components have been added, removed, or modified.

Assessment Result: The pneumatic system is outside the scope of this validation. No modifications have been made to the original pneumatic system; therefore, no additional pneumatic safety assessment or testing was required.

E10 - Maintenance Plan

Objective: To verify that the modified machine is included in the preventive maintenance program.

Assessment: The preventive maintenance plan includes the following items:

Maintenance Item	Status
X-Y table mechanical inspection	☑
Safety light curtain cleaned and visually inspected	☑
Emergency stop and light curtain function checked	☑

Evidence: Maintenance instruction in document "AA_F_212_VN_D_Maintenance-Resistance-Welding-Machine"

Assessment Result: The modified semi-automatic resistance spot welding machine has been included in the preventive maintenance program. The maintenance plan covers the X-Y positioning table and periodic functional checks of the installed safety devices to ensure continued safe operation.

E11 - Risk Assessment (ISO 12100)

Objective: To verify that the machine risk assessment has been performed in accordance with **ISO 12100 – Safety of Machinery – General Principles for Design – Risk Assessment and Risk Reduction**.

Assessment: The machine risk assessment documented in E4 has been reviewed against the methodology defined in ISO 12100.

The required information is documented in the referenced project documents as follows:

ISO 12100 Requirement	Evidence
Machine limits defined	Section 1, E5 & ASI
Intended use identified	Section 1, E5
Reasonably foreseeable misuse considered	E4, E5
Hazards identified	E4
Risks evaluated	E4
Risk reduction measures implemented	E7, E8, E15, E16
Residual risks evaluated	E4, E5, E6

Assessment Result: The machine risk assessment has been reviewed and found to be consistent with the principles of **ISO 12100**. The required machine limits / intended use / hazards / risks and risk

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reduction measures are documented in the referenced project documents. Together all these documents demonstrate compliance with the ISO 12100 risk assessment methodology.

E12 - Safety-Related Control System Review (ISO 13849-1)

Objective: To verify that the safety-related control system has been reviewed in accordance with the principles of **ISO 13849-1 – Safety of Machinery – Safety-Related Parts of Control Systems**.

Assessment: The safety-related control system of the modified semi-automatic resistance spot welding machine has been reviewed against the design principles of ISO 13849-1.

The modified semi-automatic resistance spot welding machine is not considered a high-risk machine according to Annex I of the European Machinery Regulation. The implemented safety concept has therefore been reviewed using the engineering principles of ISO 12100 and ISO 13849-1 without performing a formal Performance Level (PL) determination.

The following safety functions have been identified and implemented:

Safety Function	Implemented Components	Status
Emergency stop	Emergency stop button, Pilz PNOZ X2 safety relay, safety contactors	☑
Safety light curtain	PSEN opII4F-s-14-060 light curtain with dual OSSD outputs, Pilz PNOZ X2 safety relay	☑
Removal of power to the X-Y controller	Pilz PNOZ X2 safety relay and safety contactors	☑
Prevention of unexpected restart	Display a notification and request confirmation before proceeding with the next task.	☑

Engineering Review: The implemented safety functions have been designed using certified safety components intended for machinery applications.

The safety concept includes:

- Type 4 safety light curtain with dual OSSD outputs
- Pilz PNOZ X2 safety relay
- Safety contactors
- Manual RESET before restart
- START command required after every safety stop

The selected safety architecture is considered appropriate for the hazards identified in the machine risk assessment (E4).

Evidence:

- E4- Machine Risk Assessment
- E7- Safety Light Curtain
- E8- Electrical Safety
- E15- Emergency Stop
- [Wiring diagram Semi Automation XY Welding VN A.pdf](#)

Assessment Result: The safety-related control system has been reviewed against the principles of ISO 13849-1. Certified safety components have been selected and integrated into the machine safety

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concept. Based on the machine risk assessment, the implemented safety architecture is considered appropriate for the intended use of the machine.

A formal **Performance Level (PL)** calculation and **SISTEMA analysis** were not performed. ISO 13849-1 was used as a guideline when selecting the safety components. This document is not a declaration of full compliance with the European Machinery Regulation.

E13 - Light Curtain Fault Simulation

Objective: To verify that the machine enters a safe state when the safety light curtain detects a fault or the OSSD safety outputs are interrupted.

Test Procedure

- Simulate a light curtain fault by disconnecting one OSSD output.
- Switch on the machine.
- Attempt to start the machine.
- Restore the fault.
- Verify the normal restart procedure.

Expected Result

- Machine cannot start while the fault is present.
- Safety relay remains in the safe state.
- Safety contactors remain de-energized.
- After the fault is removed, the power must be turned off and then back on before the machine can be started again.

Assessment Result: The simulated light curtain fault was correctly detected by the safety circuit. The machine remained in a safe state and the power must be turned off and then back on before the machine can be started again.



Photo above of removed wire = disconnecting one OSSD output on the Pilz PNOZ X2.8P

Result ☒ PASS ☐ FAIL

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



E14 Power Loss & Restore

Objective: To verify that the machine behaves safely after a loss and restoration of the power supply.

Test Procedure

- Start an automatic production cycle.
- Switch off the machine power supply.
- Wait at least **10 seconds**.
- Restore the power supply.
- Observe the machine behavior.

Expected Result

- Machine stops immediately after power loss.
- No unintended movement occurs after power is restored.
- X-Y table remains stationary.
- Controller starts correctly after power is restored.
- Machine remains in a safe state.
- A new production cycle can only be started by pressing the **START** button.



Photo of the machine after power is restored

Assessment Result: The machine was tested under a complete power interruption. After the power supply was restored, the controller initialized correctly and the machine remained in a safe state. No unintended movement or automatic machine restart was observed. A new production cycle could only be started by the operator using the normal START comand.

Result ☒ **PASS** ☐ **FAIL**

E15 - Emergency Stop

Objective: To verify that the emergency stop device is correctly installed and that it performs its intended safety function.

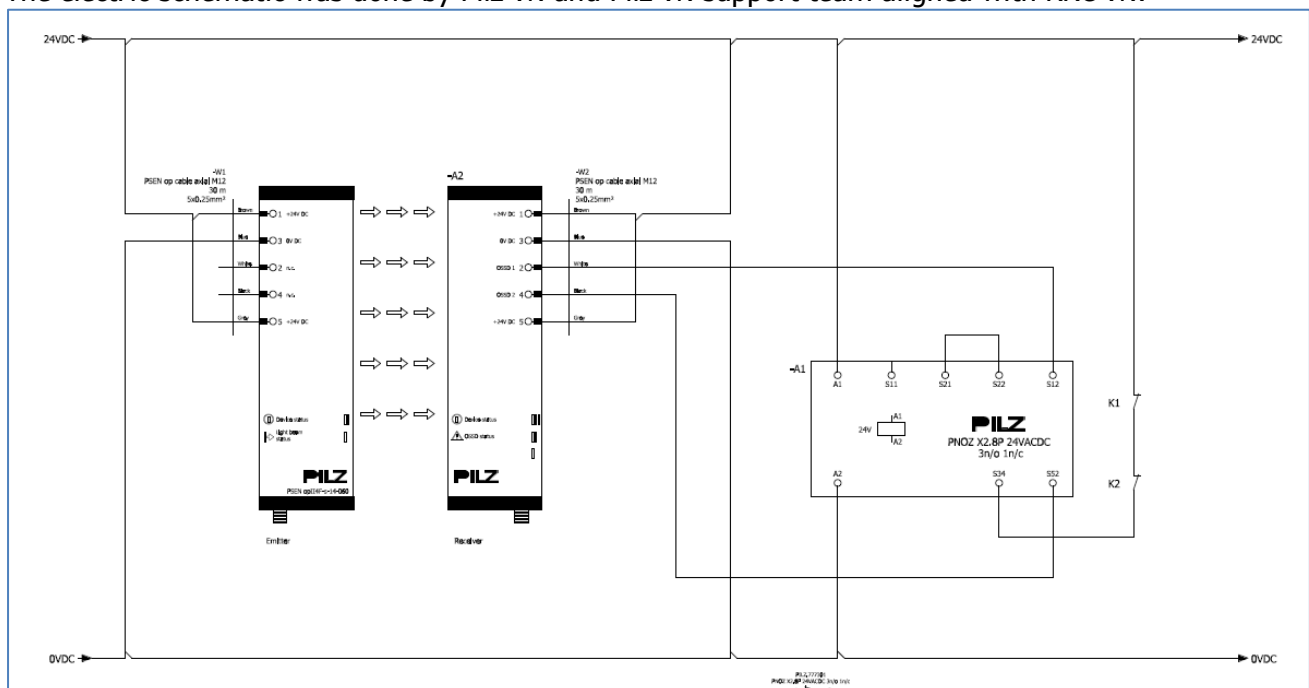
Installation Verification

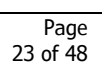
Check	Status
Emergency stop button installed	☒
Easily accessible to the operator	☒
Clearly identified	☒
Connected to the safety relay (Pilz PNOZ X2)	☒



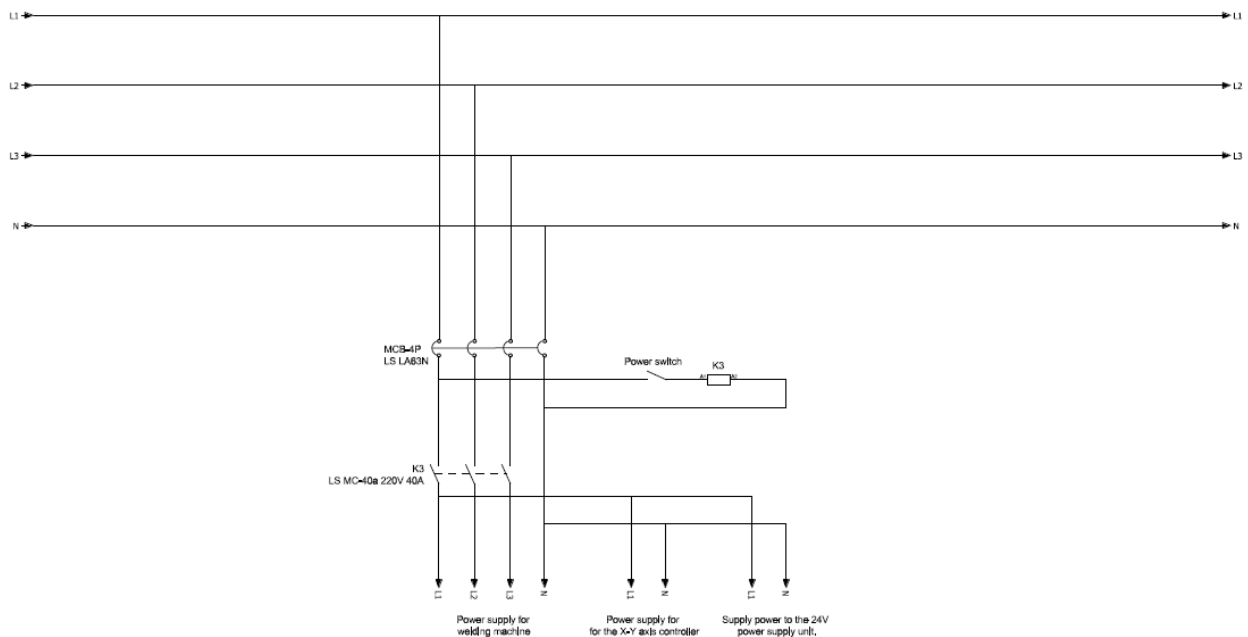
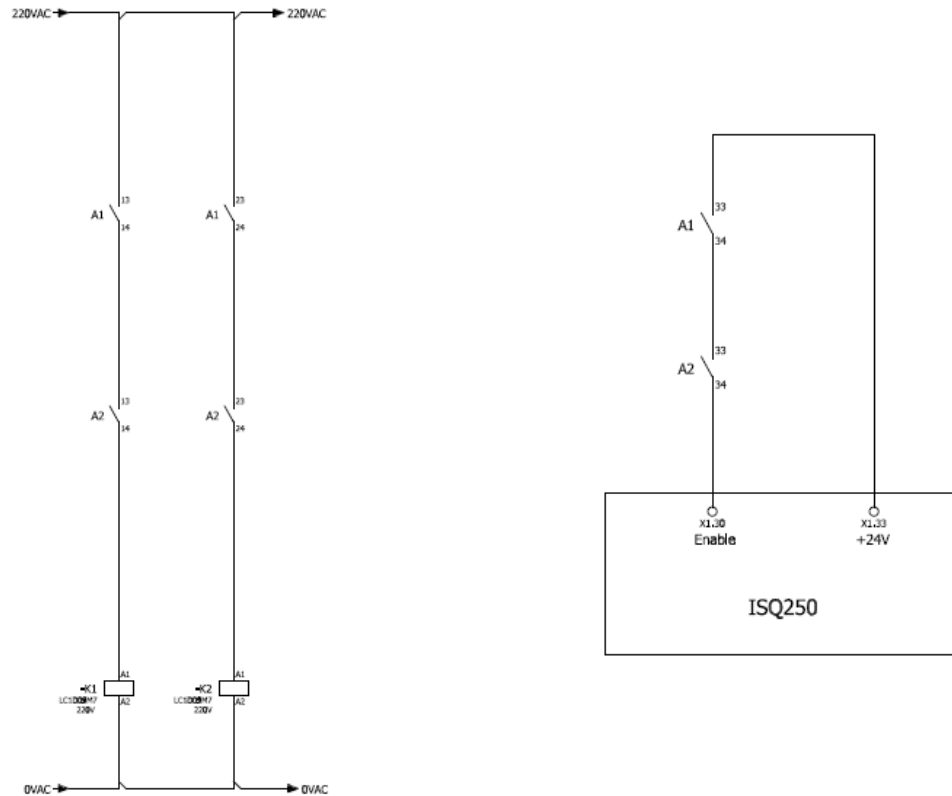
Photo of the emergency stop installation

Electrical schematic showing the emergency stop circuit as excerpt of the machine documentation. The electric schematic was done by Pilz VN and Pilz VN support team aligned with RRC VN:





Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



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Functional Test

Test	Expected Result	Result
Press emergency stop during automatic operation	Machine stops immediately	☑ PASS
	X-Y table stops	☑ PASS
	The welding machine cannot continue operating.	☑ PASS
	The pneumatically operated Z-axis remains stationary.	☑ PASS
Release emergency stop	Machine remains stopped	☑ PASS
Press START	Machine operation can only resume after completion of the defined recovery procedure and a manual START command.	☑ PASS

Assessment Result: The emergency stop has been verified for both installation and functionality. The machine immediately enters a safe state when the emergency stop is activated. Restart is only possible after the emergency stop has been released, a notification message appears on the screen requiring operator confirmation to restore the machine to the ready state, and then the Machine operation can only resume after completion of the defined recovery procedure and a manual START command..

E16 - Safety Distance Verification

Objective: To verify that the safety light curtain is installed at a sufficient separation distance from the nearest hazardous moving part of the X-Y positioning system.

Reference: ISO 13855 – Positioning of safeguards with respect to the approach of the human body.

Protective Device Specification

Item	Requirement / Value
Device type	Certified safety light curtain
Resolution / detection capability	14 mm
Safety outputs	Dual-channel OSSD1 + OSSD2
Safety relay	Pilz PNOZ X2
Installation	Vertical protective field
Protected hazard	Moving X-Y table, jig and workpiece
Planned minimum installation distance	200 mm

Remark: *The existing resistance spot welding unit, including the pneumatic Z-axis, welding head, pneumatic circuit, and associated safety functions, is part of the original CE-certified welding machine and has not been modified as part of this project. Therefore, this existing subsystem is considered an already validated machine component and is outside the scope of the stopping-time calculation for the newly added X-Y positioning table and its associated safety system.*

The safety distance is measured from the front of the light curtain to the nearest hazardous moving part of the X-Y table, jig, or workpiece.

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



Safety Distance Calculation / Response-Time Assessment

Parameter	Description	Value
T1	Safety light curtain response time acc. data sheet	9.7 ms
T2	Pilz PNOZ X2 response time acc. data sheet	15 ms
T3	Safety contactor / drive-disable response acc. data sheet	19 ms
T4	X-Y table stopping time acc. to measurement	50 ms
T	Total stopping time	93.7 ms = 0.0937 s

Safety Distance Calculation: For a 14 mm resolution safety light curtain, the additional distance related to detection capability is assumed as zero in the conventional ISO 13855 calculation.

<https://www.pilz.com/en-INT/company/news/articles/247826> Link for information only, calculation was done following ISO13855:2024

$$S = K \times T + C$$

Where:

- **S** = minimum separation distance [mm]
- **K** = human approach speed = 2000 mm/s acc. ISO13855 for “upper-limb approach”
- **T** = total stopping time = 0.0937 s
- **C** = additional distance for intrusion toward the hazard = 0 mm (0 mm is applicable for light curtains with ≤ 14 mm detection capability acc. ISO 13855)

Calculation:

$$S = 2000 \times 0.0937 + 0$$

$$S = 187.4 \text{ mm}$$

Installation Verification

Check	Requirement	Result
Calculated minimum separation distance	187.4 mm	N/A
Planned / actual installation distance	≥ 200 mm	☒ PASS
Safety margin	≥ 12.6 mm	☒ PASS
Distance measured to nearest hazardous moving part	Yes	☒ PASS
Access around / above / below the light curtain prevented	Yes	☒ PASS

Acceptance Criterion: The installation is acceptable when: **Actual separation distance ≥ calculated minimum separation distance**

Final Validation Requirement: 200 mm ≥ 187.4 mm → PASS

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



Supporting data: T4 = X-Y Table Stopping Time Measurement

Test Equipment

- Smartphone or RRC Sony Camera in slow-motion recording (approx. 240 fps)
- 24 V LED connected to the PNOZ X2 auxiliary output (NOT controller power output)

Test Procedure

- Operate the X-Y table at its maximum programmed speed.
- Start slow-motion video recording.
- Ensure the LED is visible in the recorded image together with the moving X-Y table.
- Interrupt the safety light curtain.
- The LED turning OFF indicates that the safety relay output has switched. The safety circuit then removes power from the X-Y motor driver stages while the X-Y motion controller remains powered. Review the video frame by frame.
- Measure the time from **LED OFF** until the X-Y table has completely stopped.
- Repeat the measurement **5 times**.
- Record all measured values and use the **maximum measured stopping time** for the safety distance calculation.

T4 = X-Y Table Stopping Time = MAX (50, 50, 49, 49, 50) = 50ms

E17 - Workpiece Preparation & Fixture Validation

Objective: To verify that the modified fixture design and clamping concept provide repeatable and correct positioning of the battery pack and cell tabs before automatic welding. The validation demonstrates that the transition from the manual welding process to the semi-automatic welding process maintains the required product quality by ensuring consistent workpiece preparation prior to welding.

Background: In the previous manual welding process, the operator visually checked the position of each cell tab and corrected minor misalignments before every welding operation.

With the introduction of the semi-automatic welding process the fixture has been redesigned to accurately position and clamp the battery pack and the cell tabs before welding starts. Correct positioning is therefore achieved by the fixture design instead of manual adjustment by the operator.

Incorrect battery loading is prevented during the workpiece preparation process before the battery enters the automatic welding station and is therefore outside the scope of this validation.

Missing cell tabs are not introduced by the modified process. Detection of missing or defective weld conditions remains part of the welding monitor and is already validated in the previous manual welding.

Test	Expected Result (Evidence)	Result
Correct workpiece preparation	The battery pack is correctly positioned in the fixture. All cell tabs are correctly aligned and securely clamped before welding. Figure E17-1 documents the correctly prepared workpiece.	☒ PASS

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine

Misaligned cell tab (Challenge Test)	An intentionally misaligned cell tab cannot be correctly clamped and the incorrect condition is clearly visible before welding. Figure E17-2 documents the challenge test.	☒ PASS
Fixture repeatability	The same battery pack is loaded and unloaded five consecutive times. After each loading cycle, the position of every cell tab is measured using the optical measuring system. The measured X and Y deviations relative to the fixture reference are documented in Table E17-1 . Figure E17-3 shows the measurement setup.	☒ PASS

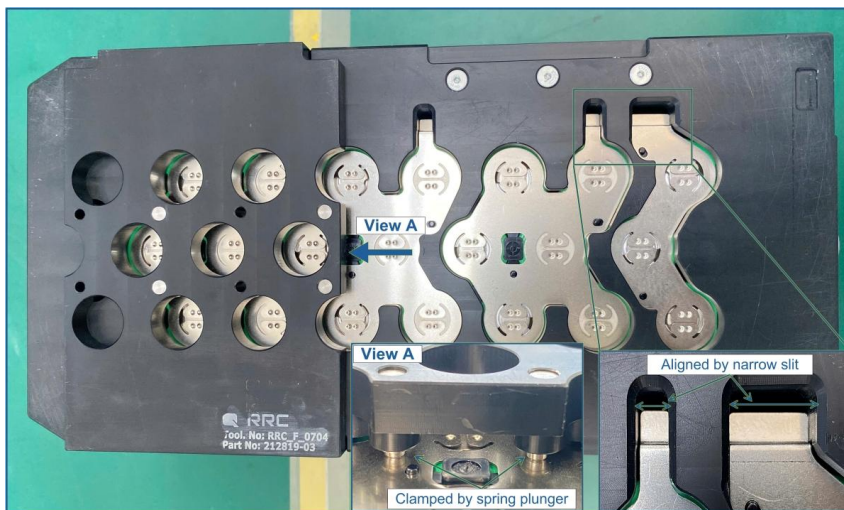


Figure E17-1 – Correctly positioned battery pack with all cell tabs correctly aligned and clamped.

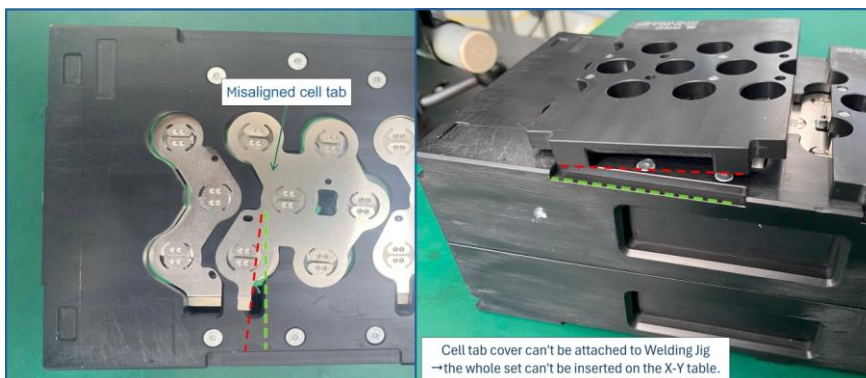


Figure E17-2 – Challenge test with an intentionally misaligned cell tab demonstrating that incorrect positioning cannot be accepted for welding.



Figure E17-3 – Optical measurement setup used to verify repeatable positioning of the battery pack and cell tabs.

Table E17-1 – Optical Measurement Report (5 repeated loading cycles) with Fixture ID = RRC_F_0702, RRC_F_0703, RRC_F_0704, RRC_F_0705, RRC_F_0722, RRC_F_0723 on measuring device Manual Video Measuring / Q-ID: RRC_Q_0045 ... measuring results below in unit mm

Cell Tab No. (side A / B)	Nominal X	Nominal Y	Run 1 ΔX	Run 1 ΔY	Run 2 ΔX	Run 2 ΔY	Run 3 ΔX	Run 3 ΔY	Run 4 ΔX	Run 4 ΔY	Run 5 ΔX	Run 5 ΔY	Max ΔX	Max ΔY	Result OK / NOK
1A	48.9238	29.6887	0.1197	0.0783	0.1131	0.1173	0.0825	0.0909	0.1391	0.0781	0.0649	0.1130	0.1391	0.1173	OK
2A	72.4340	30.1438	0.0519	0.1665	0.1254	0.1378	0.0538	0.1150	0.0954	0.1270	0.0715	0.1743	0.1254	0.1743	OK
3A	95.8992	30.6380	0.0781	0.0756	0.0774	0.1720	0.0648	0.1668	0.1001	0.1477	0.1378	0.1795	0.1378	0.1795	OK
4A	119.3606	31.1845	0.1112	0.1432	0.0715	0.0950	0.0509	0.0986	0.1390	0.0743	0.1396	0.1049	0.1396	0.1432	OK
5A	142.8063	31.7573	0.1458	0.0730	0.0784	0.1095	0.0600	0.1724	0.1324	0.1772	0.1272	0.1316	0.1458	0.1772	OK
6A	166.2796	32.2742	0.0524	0.1675	0.1008	0.1612	0.1391	0.1159	0.0702	0.0775	0.1178	0.1568	0.1391	0.1675	OK
7A	189.7316	32.8308	0.1220	0.1604	0.1440	0.0874	0.0642	0.1464	0.0780	0.1018	0.1026	0.1065	0.1440	0.1604	OK
8A	177.5082	52.8589	0.1081	0.1520	0.0570	0.1611	0.0784	0.1631	0.1432	0.0758	0.0955	0.0748	0.1432	0.1631	OK

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9A	154.0683	52.2849	0.0622	0.0802	0.0740	0.1461	0.1193	0.1638	0.1339	0.0890	0.0643	0.1302	0.1339	0.1638	OK
10A	130.5725	51.7747	0.0513	0.0714	0.1393	0.0914	0.0922	0.1712	0.0543	0.0847	0.1380	0.0729	0.1393	0.1712	OK
11A	107.1458	51.2081	0.1325	0.1149	0.0946	0.1427	0.1366	0.1045	0.0684	0.1540	0.0578	0.1469	0.1366	0.1540	OK
12A	83.7177	50.6909	0.1112	0.1436	0.0767	0.1202	0.1160	0.1431	0.1040	0.0796	0.0836	0.0863	0.1160	0.1436	OK
13A	60.2790	50.1754	0.1004	0.1410	0.0627	0.1777	0.0846	0.1465	0.0897	0.0832	0.0943	0.1181	0.1004	0.1777	OK
14A	36.7486	49.7386	0.0953	0.1017	0.1238	0.1150	0.1013	0.1117	0.1324	0.1484	0.1357	0.1659	0.1357	0.1659	OK
15A	48.0362	70.3081	0.0530	0.1301	0.1419	0.1596	0.1137	0.1370	0.0772	0.1266	0.0792	0.0860	0.1419	0.1596	OK
16A	71.5529	70.7333	0.1483	0.1324	0.1029	0.1327	0.1373	0.1371	0.0715	0.1247	0.0629	0.1527	0.1483	0.1527	OK
17A	94.9944	71.2581	0.1314	0.1735	0.0700	0.0854	0.1062	0.1089	0.0521	0.0842	0.1226	0.1618	0.1314	0.1735	OK
18A	118.3843	71.8073	0.1427	0.0935	0.0714	0.1418	0.1219	0.0990	0.1398	0.1058	0.0729	0.1799	0.1427	0.1799	OK
19A	141.8083	72.3671	0.1457	0.1343	0.0997	0.0774	0.0860	0.1224	0.0627	0.0731	0.1399	0.1715	0.1457	0.1715	OK
20A	165.2977	72.8730	0.1309	0.0880	0.1401	0.1531	0.0896	0.1544	0.0892	0.1296	0.0972	0.1461	0.1401	0.1544	OK
21A	188.7334	73.4469	0.0601	0.1203	0.0676	0.1457	0.0538	0.0955	0.1451	0.1747	0.0838	0.0722	0.1451	0.1747	OK
1B	49.6415	28.4301	0.0952	0.1164	0.0960	0.1730	0.0545	0.1724	0.0985	0.1371	0.1227	0.1527	0.1227	0.1730	OK
2B	73.0805	28.4950	0.0545	0.1145	0.0618	0.1512	0.0778	0.1367	0.1347	0.1466	0.0971	0.1439	0.1347	0.1512	OK
3B	96.5419	28.6502	0.1267	0.0796	0.1484	0.0794	0.1186	0.1624	0.1007	0.1569	0.0918	0.1615	0.1484	0.1624	OK
4B	119.9675	28.7393	0.1034	0.0719	0.1002	0.1131	0.0704	0.0932	0.1401	0.1348	0.1429	0.1301	0.1429	0.1348	OK
5B	143.1797	28.7801	0.1345	0.0836	0.0635	0.1428	0.1038	0.0955	0.0889	0.1611	0.1269	0.1435	0.1345	0.1611	OK
6B	166.6185	28.7499	0.0622	0.1098	0.1498	0.1069	0.0703	0.1608	0.0973	0.1473	0.0803	0.1282	0.1498	0.1608	OK
7B	190.2471	28.9919	0.1453	0.1256	0.0917	0.1447	0.0880	0.1151	0.0568	0.1573	0.1136	0.0933	0.1453	0.1573	OK
8B	178.6022	49.3523	0.1345	0.1202	0.1117	0.1789	0.0826	0.1098	0.0616	0.1673	0.1032	0.0852	0.1345	0.1789	OK
9B	154.9292	49.0864	0.1185	0.1364	0.1492	0.1742	0.0763	0.0773	0.0689	0.1228	0.1358	0.0781	0.1492	0.1742	OK
10B	131.4872	49.1196	0.1336	0.1432	0.0701	0.1062	0.0905	0.1731	0.1437	0.0930	0.0592	0.1761	0.1437	0.1761	OK
11B	108.1733	49.0102	0.1136	0.1766	0.0758	0.1009	0.1348	0.1506	0.0624	0.1580	0.0987	0.1588	0.1348	0.1766	OK
12B	84.7250	48.9163	0.1305	0.0727	0.1214	0.0701	0.1495	0.1613	0.0505	0.0929	0.1182	0.1759	0.1495	0.1759	OK
13B	61.3142	48.7883	0.1374	0.1520	0.0640	0.1618	0.1366	0.0809	0.1354	0.0831	0.1157	0.1562	0.1374	0.1618	OK
14B	37.8658	48.7298	0.0670	0.0868	0.1265	0.1609	0.1163	0.0782	0.1351	0.1758	0.0824	0.1319	0.1351	0.1758	OK
15B	49.5420	69.0609	0.1185	0.1729	0.0674	0.1041	0.1492	0.1645	0.0738	0.1675	0.1459	0.1243	0.1492	0.1729	OK
16B	72.9804	69.1229	0.1181	0.1363	0.1096	0.1096	0.1432	0.1144	0.1256	0.0926	0.0833	0.1789	0.1432	0.1789	OK
17B	96.3598	69.2712	0.0677	0.1577	0.0815	0.1656	0.1028	0.1762	0.1225	0.1173	0.0821	0.0944	0.1225	0.1762	OK
18B	119.7945	69.3671	0.1441	0.1179	0.0765	0.1495	0.0805	0.1590	0.0543	0.1694	0.0500	0.0780	0.1441	0.1694	OK
19B	143.2281	69.4097	0.0694	0.1010	0.0583	0.0705	0.0804	0.1206	0.1445	0.1137	0.0734	0.0850	0.1445	0.1206	OK
20B	166.6698	69.3925	0.1033	0.1232	0.1320	0.1621	0.1157	0.1414	0.1283	0.1774	0.1158	0.1138	0.1320	0.1774	OK
21B	190.3914	69.6246	0.0702	0.0721	0.1253	0.1764	0.1025	0.1166	0.1188	0.1626	0.0917	0.1339	0.1253	0.1764	OK

Acceptance Criteria:

- The fixture shall provide repeatable positioning of the battery pack.
- All cell tabs shall be correctly aligned and securely clamped before welding.
- An intentionally misaligned cell tab shall be clearly visible and shall not be accepted for production.
- The measured X and Y deviations after five consecutive loading cycles shall remain within the specified drawing tolerance.

Assessment Result: The modified fixture and clamping concept were verified to provide repeatable positioning of the battery pack and cell tabs before automatic welding. The validation confirmed that the redesigned fixture replaces the manual alignment process used during manual welding and

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provides a controlled and repeatable workpiece preparation process. The repeatability measurements demonstrated consistent positioning of the battery pack and cell tabs throughout repeated loading cycles.

E18 – Welding Position & Process Parameter Verification

Objective: To verify that the automatic welding system positions the welding head at the correct programmed welding locations and uses the correct approved welding program for every welding operation.

The validation confirms that each programmed welding position is accurately reached by the X-Y positioning system and that the corresponding welding program displayed by the welding machine matches the approved welding specification.

Background: The battery pack is welded from two sides. Each side consists of **21 programmed welding positions**, resulting in a total of **42 welding operations**. For every welding position, the control software automatically moves the X-Y positioning table to the programmed location and selects the corresponding welding program on the welding machine.

Test	Expected Result (Evidence)	Result
Welding Position Verification	The welding electrodes are temporarily replaced by two fine-tip markers. A flat reference block with paper is installed in the fixture and the automatic cycle is executed with welding disabled. The resulting 21 marker positions are measured using the optical measuring system and compared with the nominal positions. The measured X and Y deviations are documented in Table E18-1 . The same verification is valid for Side B after the fixture is turned.	☒ PASS
Welding Program Verification	During execution of the automatic welding sequence, the welding program displayed on the welding machine is verified for every welding position on Side A and Side B. The displayed welding program shall match the approved welding specification. Results are documented in Table E18-2 .	☒ PASS



Figure E18-1 – Welding position verification test setup with fine-tip markers.

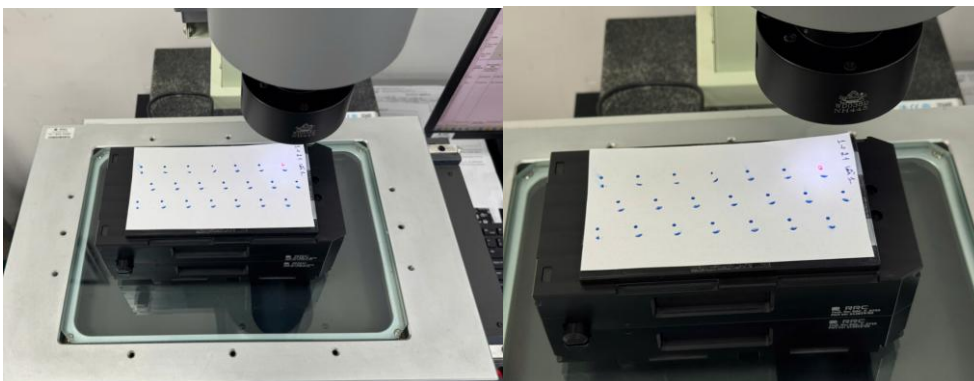


Figure E18-2 – Optical measurement of the marker positions.

Table E18-1 – Welding position measurement results.

Pos.	Nominal X (mm)	Nominal Y (mm)	Measured X (mm)	Measured Y (mm)	ΔX (mm)	ΔY (mm)	Tolerance (mm)	Result
01	166.4	160.075	166.400	160.075	0	0	0.0000	☑ PASS
02	166.4	183.525	166.317	183.566	0.0827	0.0407	0.0922	☑ PASS
03	166.4	206.975	166.331	207.072	0.0688	0.0967	0.1187	☑ PASS
04	166.4	230.425	166.363	230.362	0.037	0.0627	0.0728	☑ PASS
05	166.4	253.875	166.226	254.064	0.1741	0.1891	0.2570	☑ PASS
06	166.4	277.325	166.223	277.486	0.1773	0.1612	0.2396	☑ PASS

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07	166.4	300.775	166.256	300.886	0.1438	0.1106	0.1814	☒ PASS
08	146.09	289.055	146.262	289.150	0.1715	0.0951	0.1961	☒ PASS
09	146.09	265.605	146.200	265.688	0.11	0.0833	0.1380	☒ PASS
10	146.09	242.155	146.217	242.243	0.1274	0.0878	0.1547	☒ PASS
11	146.09	218.705	146.177	218.751	0.0874	0.046	0.0988	☒ PASS
12	146.09	195.255	146.181	195.325	0.091	0.0699	0.1147	☒ PASS
13	146.09	171.805	146.158	171.840	0.0676	0.0348	0.0760	☒ PASS
14	146.09	171.705	146.125	171.719	0.0353	0.0141	0.0380	☒ PASS
15	125.78	160.075	125.786	160.139	0.0062	0.0635	0.0638	☒ PASS
16	125.78	183.525	125.874	183.579	0.0942	0.0539	0.1085	☒ PASS
17	125.78	206.975	125.874	207.047	0.0935	0.0715	0.1177	☒ PASS
18	125.78	230.425	125.902	230.533	0.1215	0.1081	0.1626	☒ PASS
19	125.78	253.875	125.858	253.995	0.0779	0.1198	0.1429	☒ PASS
20	125.78	277.325	125.908	277.454	0.1278	0.1288	0.1814	☒ PASS
21	125.78	300.775	125.904	300.841	0.1236	0.0656	0.1399	☒ PASS

Summary

Parameter	Result
Maximum ΔX	0.1773
Maximum ΔY	0.1891
All positions within tolerance ($\pm 0.3\text{mm}$)	☒ YES ☐ NO
Overall Result	☒ PASS ☐ FAIL

Table E18-2 – Welding program verification.

Position	Side A Required Program	Side A Displayed Program	Side A Result	Side B Required Program	Side B Displayed Program	Side B Result
01	01	01	☒ PASS	22	22	☒ PASS
02	02	02	☒ PASS	23	23	☒ PASS
03	03	03	☒ PASS	24	24	☒ PASS
04	04	04	☒ PASS	25	25	☒ PASS
05	05	05	☒ PASS	26	26	☒ PASS
06	06	06	☒ PASS	27	27	☒ PASS
07	07	07	☒ PASS	28	28	☒ PASS
08	08	08	☒ PASS	29	29	☒ PASS
09	09	09	☒ PASS	30	30	☒ PASS
10	10	10	☒ PASS	31	31	☒ PASS
11	11	11	☒ PASS	32	32	☒ PASS
12	12	12	☒ PASS	33	33	☒ PASS
13	13	13	☒ PASS	34	34	☒ PASS
14	14	14	☒ PASS	35	35	☒ PASS
15	15	15	☒ PASS	36	36	☒ PASS
16	16	16	☒ PASS	37	37	☒ PASS

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



17	17	17	☒ PASS	38	38	☒ PASS
18	18	18	☒ PASS	39	39	☒ PASS
19	19	19	☒ PASS	40	40	☒ PASS
20	20	20	☒ PASS	41	41	☒ PASS
21	21	21	☒ PASS	42	42	☒ PASS

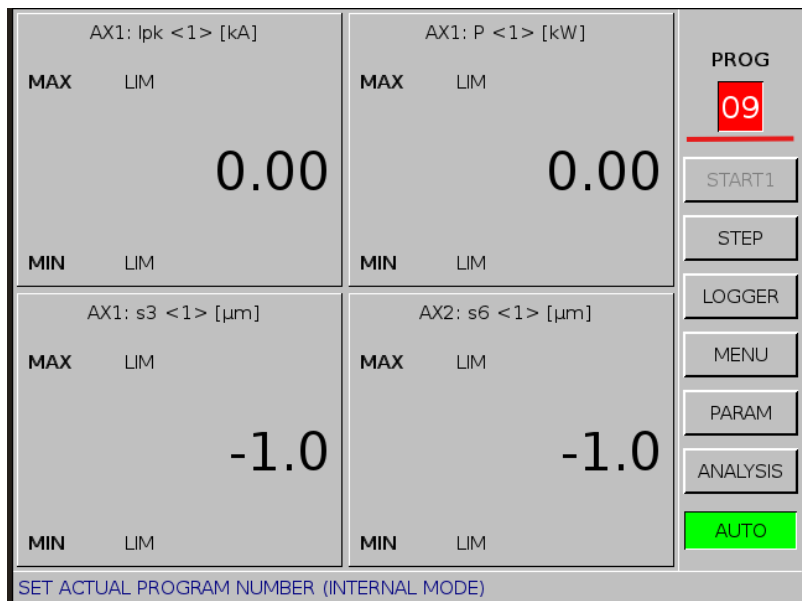


Figure E18-3 – Example of welding program displayed on the welding machine.

Acceptance Criteria:

- All measured welding positions shall be within the specified positional tolerance.
- The displayed welding program shall correspond to the approved welding specification for every welding position.
- No incorrect welding program shall be accepted.

Assessment Result: The validation confirmed that the X-Y positioning system accurately reaches all programmed welding positions and that the welding machine selects the correct welding program for every welding operation.

E19 – Welding Result & NOK Handling

Objective: To verify that the automatic welding system correctly detects non-conforming welding results and prevents further processing of a potentially non-conforming battery pack.

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



The validation confirms that a welding fault immediately stops the automatic welding process, locks the machine, records the event, and requires authorized supervisor intervention before production can continue.

Background: After each welding operation, the welding machine evaluates the welding result using the integrated welding monitor.

If a welding result is outside the specified process limits, the machine shall immediately stop the automatic welding sequence. The battery pack shall remain under controlled status until it has been reviewed by the Production Team Leader.

The complete event shall be recorded to ensure full product traceability.

Test	Expected Result	Result
Normal welding result	A welding result within the specified limits is accepted as OK and the automatic welding sequence continues with the next welding position. Table E19-1.	☒ PASS
Non-conforming welding result (NOK)	A simulated or actual welding result outside the specified limits immediately stops the automatic welding sequence. The machine enters the locked state and no additional welding operation is possible. The NOK event is recorded together with the Work Order, UDI/Serial Number, welding position, date/time and error information. Table E19-1 and Figure E19-1.	☒ PASS
Supervisor release	The machine can only be unlocked by authorized Production Team Leader personnel using a password-protected release . Before welding can continue the correct restart welding position must be confirmed. The supervisor release and restart position are recorded in the production log. Table E19-2.	☒ PASS

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine

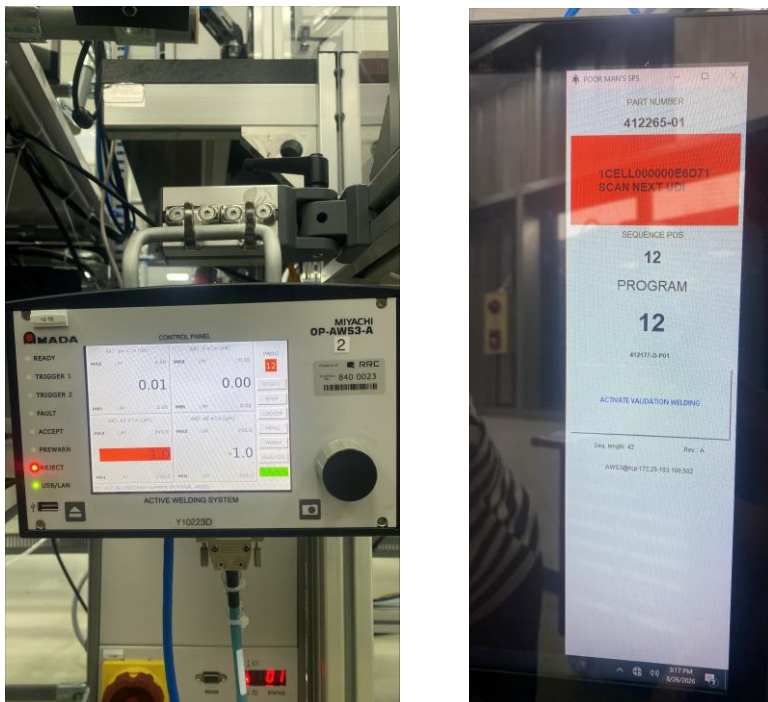


Figure E19-1 – Machine status after welding NOK: left side Welding Monitor & Right side Soft PLC “SPS”

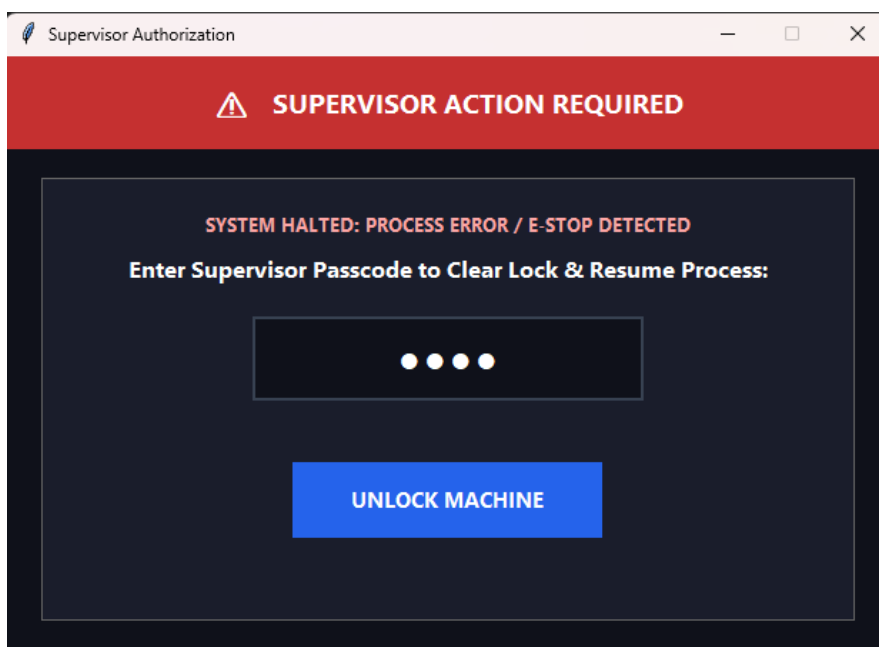


Figure E19-2 – Password-protected supervisor release.

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



```
Python_Libs > rrc > production_events_line3_manual.csv
386 2026-08-26 15:25:59.421,20260826_152524_ee628a,,1CELL000000E6D71,MOVE_STARTED,StageStates.PR
387 2026-08-26 15:26:00.546,20260826_152524_ee628a,,1CELL000000E6D71,POSITION_VERIFIED,StageStates.PROCESS_COMMAND,,146.09,289.055,146.0875,289.05,PASS,
388 2026-08-26 15:26:00.673,20260826_152524_ee628a,,1CELL000000E6D71,PROGRAM_SELECTION_STARTED,StageStates.PROCESS_COMMAND,8,,,,,8,,,,,Welding program
389 2026-08-26 15:26:00.773,20260826_152524_ee628a,,1CELL000000E6D71,PROGRAM_VERIFIED,StageStates.PROCESS_COMMAND,8,,,,,8,,,,,Welding program was ver
390 2026-08-26 15:26:00.773,20260826_152524_ee628a,,1CELL000000E6D71,WELD_STARTED,StageStates.PROCESS_COMMAND,8,146.09,289.055,,8,8,,,,,Start the weld
391 2026-08-26 15:26:02.453,20260826_152524_ee628a,,1CELL000000E6D71,WELD_RESULT,StageStates.CHECK_WELD_RESULT,8,,,,,8,8,OK,,,Welding completed succes
392 2026-08-26 15:26:02.848,20260826_152524_ee628a,,1CELL000000E6D71,START_BUTTON_PRESSED,StageStates.START,,,,,,START button pressed,,Operator pre
393 2026-08-26 15:26:03.185,20260826_152524_ee628a,,1CELL000000E6D71,MOVE_STARTED,StageStates.PROCESS_COMMAND,,146.09,265.605,,,,,,Xytable starts mo
394 2026-08-26 15:26:03.859,20260826_152524_ee628a,,1CELL000000E6D71,POSITION_VERIFIED,StageStates.PROCESS_COMMAND,,146.09,265.605,146.0875,265.6,PASS,,
395 2026-08-26 15:26:03.975,20260826_152524_ee628a,,1CELL000000E6D71,PROGRAM_SELECTION_STARTED,StageStates.PROCESS_COMMAND,9,,,,,9,,,,,Welding program
396 2026-08-26 15:26:04.082,20260826_152524_ee628a,,1CELL000000E6D71,PROGRAM_VERIFIED,StageStates.PROCESS_COMMAND,9,,,,,9,,,,,Welding program was ver
397 2026-08-26 15:26:04.082,20260826_152524_ee628a,,1CELL000000E6D71,WELD_STARTED,StageStates.PROCESS_COMMAND,9,146.09,265.605,,9,9,,,,,Start the weld
398 2026-08-26 15:26:05.630,20260826_152524_ee628a,,1CELL000000E6D71,WELD_RESULT,StageStates.CHECK_WELD_RESULT,9,,,,,9,9,OK,,,Welding completed succes
399 2026-08-26 15:26:06.004,20260826_152524_ee628a,,1CELL000000E6D71,START_BUTTON_PRESSED,StageStates.START,,,,,,START button pressed,,Operator pre
400 2026-08-26 15:26:06.345,20260826_152524_ee628a,,1CELL000000E6D71,MOVE_STARTED,StageStates.PROCESS_COMMAND,,146.09,242.155,,,,,,Xytable starts mo
401 2026-08-26 15:26:07.018,20260826_152524_ee628a,,1CELL000000E6D71,POSITION_VERIFIED,StageStates.PROCESS_COMMAND,,146.09,242.155,146.0875,242.15,PASS,,
402 2026-08-26 15:26:07.134,20260826_152524_ee628a,,1CELL000000E6D71,PROGRAM_SELECTION_STARTED,StageStates.PROCESS_COMMAND,10,,,,,10,,,,,Welding progr
403 2026-08-26 15:26:07.233,20260826_152524_ee628a,,1CELL000000E6D71,PROGRAM_VERIFIED,StageStates.PROCESS_COMMAND,10,,,,,10,10,,,,,Welding program was
404 2026-08-26 15:26:07.274,20260826_152524_ee628a,,1CELL000000E6D71,WELD_STARTED,StageStates.PROCESS_COMMAND,10,146.09,242.155,,10,10,,,,,Start the w
405 2026-08-26 15:26:08.779,20260826_152524_ee628a,,1CELL000000E6D71,WELD_RESULT,StageStates.CHECK_WELD_RESULT,10,,,,,10,10,NOK,Welding reject signal,,
406 2026-08-26 15:26:08.798,20260826_152524_ee628a,,1CELL000000E6D71,SUPERVISOR_AUTHORIZATION_REQUESTED,StageStates.ERROR_WELD,10,,,,,,Welding failur
407
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Move to WELD command!!!
Motion Controller Connected2: True
Position safe. Enabling foot pedal.
Welding position 10.
Wait until welding is done.
Machine is ready: False
Welding result: {'ready': 0, 'operational_mode': 0, 'reject': 1, 'hfi_device_fault': 0, 'ok': 0}
Move to state 5
Welding disabled for transit.
Welding failed!
ADAM DI 6 weld result failed!
Move to state 7
Error during welding process! Please check the machine and try again.
Waiting for supervisor to input a PASSCODE and unlock the machine...

```

Figure E19-3 – Production log showing recorded NOK event.

Acceptance Criteria:

- Every welding result outside the specified limits shall immediately stop the automatic welding process.
- The machine shall automatically enter the locked state.
- Additional welding shall not be possible until authorized release.
- Machine release shall require password authentication by the Production Team Leader.
- Before restarting, the correct welding position shall be confirmed by the Production Team Leader.
- The complete NOK event shall be stored together with:
 - Work Order
 - UDI / Serial Number
 - Welding position
 - Welding program
 - Date and time
 - Welding result / error description
 - Supervisor release
 - Restart welding position

Assessment Result: The validation confirmed that the automatic welding system reliably detects non-conforming welding results and immediately prevents further processing of the battery pack. The machine entered the locked state, required password-protected supervisor release, and recorded the complete event in the production log. The controlled release procedure ensured that production could only continue after verification of the correct restart welding position.

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



Table E19-1 – Welding Result Verification

Verification Item	Expected Result	Verified	Result
Welding result within limits	Automatic sequence continues	☒	☒ PASS
Welding result outside limits	Automatic sequence stops immediately	☒	☒ PASS
Machine enters locked state	YES	☒	☒ PASS
Additional welding prevented	YES	☒	☒ PASS
NOK event recorded	YES	☒	☒ PASS

Table E19-2 – Supervisor Release Verification

Verification Item	Expected Result	Verified	Result
Password required	YES	☒	☒ PASS
Release restricted to Production Team Leader	YES	☒	☒ PASS
Restart welding position confirmation required	YES	☒	☒ PASS
Supervisor release recorded in production log	YES	☒	☒ PASS
Production continues only after successful release	YES	☒	☒ PASS

Figure E19-3 = log entry.

Date/Time	WO	UDI	Weld Position	Program	Event	User	Action
2026-08-15 09:41:12	10012 34	B000123 45	17	P05	Weld Out of Limits	System	Machine Locked
2026-08-15 09:45:08	10012 34	B000123 45	17	P05	Supervisor Release	PTL	Restart Position Confirmed

E20 – Process Interruption & Controlled Recovery

Objective: To verify that the semi-automatic welding process safely handles process interruptions and allows controlled recovery without affecting product quality. The validation confirms that process interruptions do not result in skipped welds, duplicate welds or incorrect continuation of the welding sequence.

Background: During automatic operation, a battery pack may already be partially welded when the welding process is interrupted.

Following a process interruption, the battery pack cannot simply be restarted automatically. The Production Team Leader or other supervisor must inspect the battery pack, determine the correct restart position and release the machine before production can continue.

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



This controlled recovery process ensures that the remaining welding operations are completed correctly and that product quality is maintained.

Validation:

Test	Expected Result (Evidence)	Result
Emergency Stop	Pressing the Emergency Stop immediately stops the automatic welding process. After reset, automatic restart is not possible. Production can only continue using the defined recovery procedure. Table E20-1.	☒ PASS
Safety Light Curtain	Interrupting the safety light curtain immediately stops the automatic welding process. After reset, automatic restart is not possible. Production can only continue using the defined recovery procedure. Table E20-1.	☒ PASS
Power Failure	Simulated power failure immediately stops the welding process. After power restoration, production cannot continue automatically and requires the defined recovery procedure. Table E20-1.	☒ PASS
Controlled Recovery	The Production Team Leader verifies the battery pack, selects the correct restart welding position and resumes production. Welding continues from the selected position without skipping or duplicate welds. Table E20-2.	☒ PASS

Evidence:



Figure E20-1 – Emergency Stop during automatic welding: Emergency Deactivated

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



Figure E20-1 – Emergency Stop during automatic welding: Emergency: Activated



Figure E20-2 – Safety light curtain interruption: Safety light curtain Deactivated

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



Figure E20-2 – Safety light curtain interruption: Safety light curtain Activated



Figure E20-3 – Power failure recovery: Machine Power On after Power Failure

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine

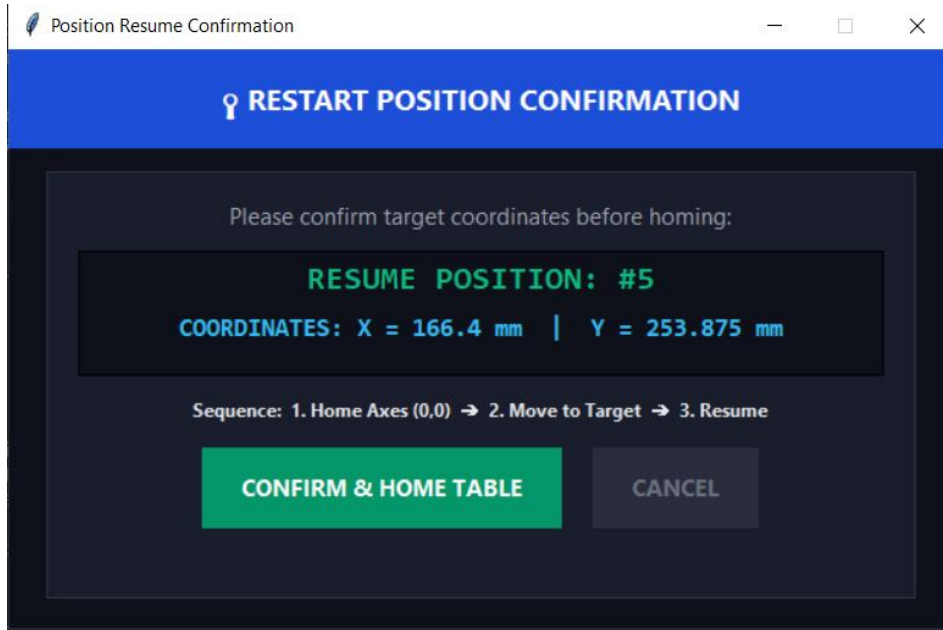


Figure E20-4 – Restart welding position selection

Table E20-1 – Process interruption verification

Interruption Scenario	Process Stopped	Automatic Restart Prevented	Recovery Procedure Required	Result
Emergency Stop	☒	☒	☒	☒ PASS
Safety Light Curtain	☒	☒	☒	☒ PASS
Power Failure	☒	☒	☒	☒ PASS

Table E20-2 – Controlled recovery verification

Verification Item	Expected Result	Verified	Result
Battery pack inspected by Production Team Leader	YES	☒	☒ PASS
Correct restart welding position selected	YES	☒	☒ PASS
Welding resumed from selected position	YES	☒	☒ PASS
No welding position skipped	YES	☒	☒ PASS
No welding position repeated unintentionally	YES	☒	☒ PASS

Acceptance Criteria

- Emergency Stop shall immediately stop the automatic welding process.
- Safety light curtain activation shall immediately stop the automatic welding process.
- Power failure shall immediately stop the automatic welding process.

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



- Automatic continuation of the welding sequence shall not be possible after any interruption.
- The Production Team Leader shall inspect the battery pack before production resumes.
- The correct restart welding position shall be selected before continuing the welding process.
- The welding sequence shall continue from the selected position without skipped or duplicate welds.

Assessment Result: The validation confirmed that all defined interruption scenarios safely stopped the automatic welding process. Production could only continue after the battery pack had been inspected and the correct restart welding position had been selected. The recovery procedure ensured that the welding sequence was completed without skipping or duplicate welds.

E21– Product Traceability & Welding Data Recording

Objective: To verify that the semi-automatic welding process records the Work Order, UDI / Serial Number and the related welding data for every battery pack.

The same Python library and data recording method used for the existing manual welding process are also used for the semi-automatic welding process.

Test	Expected Result (Evidence)	Result
WO, UDI and welding data recording	The production record contains the correct Work Order and UDI / Serial Number together with the related welding data. Welding data are stored for both OK and NOK welding results. Figure E21-1 shows the recorded production data.	☑ PASS

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



Timestamp	Cycle_ID	Work_Order	UDI	Start_Type	Welding_Position	Target_X	Target_Y	Actual_X	Actual_Y	Position_Result	Required_Program	Actual_Program	Weld_Result	Details
27/08/2026 15:38:54	20260827_153648_95690		1003195	SESSION_START										XY-table automation script started
27/08/2026 15:39:03	20260827_153648_95690		1003195	CYCLE_STARTED										New battery welding cycle started
27/08/2026 15:39:03	20260827_153648_95690		1003195	UDI_SCANNED										UDI=1CELL000000E071
27/08/2026 15:39:36	20260827_153648_95690		1003195	START_BUTTON_PRESSED										Operator pressed the Start button
27/08/2026 15:39:37	20260827_153648_95690		1003195	MOVE_STARTED		166.4	160.075							XYtable starts moving to the next welding position
27/08/2026 15:39:38	20260827_153648_95690		1003195	POSITION_VERIFIED		166.4	160.075	166.4	160.075	PASS				XYtable reached the target welding position
27/08/2026 15:39:44	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	1						1			Welding program was set on the welding machine
27/08/2026 15:39:42	20260827_153648_95690		1003195	PROGRAM_VERIFIED	1						1		1	Welding program was verified before actual welding
27/08/2026 15:39:42	20260827_153648_95690		1003195	WELD_STARTED	1	166.4	160.075							Start the welding process
27/08/2026 15:39:43	20260827_153648_95690		1003195	WELD_RESULT	1						1		1 OK	Welding completed successfully
27/08/2026 15:39:44	20260827_153648_95690		1003195	MOVE_STARTED		166.4	183.525							XYtable starts moving to the next welding position
27/08/2026 15:39:44	20260827_153648_95690		1003195	POSITION_VERIFIED		166.4	183.525	166.4	183.525	PASS				XYtable reached the target welding position
27/08/2026 15:39:44	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	2						2			Welding program was set on the welding machine
27/08/2026 15:39:45	20260827_153648_95690		1003195	PROGRAM_VERIFIED	2						2		2	Welding program was verified before actual welding
27/08/2026 15:39:45	20260827_153648_95690		1003195	WELD_STARTED	2	166.4	183.525							Start the welding process
27/08/2026 15:39:46	20260827_153648_95690		1003195	WELD_RESULT	2						2		2 OK	Welding completed successfully
27/08/2026 15:39:46	20260827_153648_95690		1003195	MOVE_STARTED		166.4	206.975							XYtable starts moving to the next welding position
27/08/2026 15:39:47	20260827_153648_95690		1003195	POSITION_VERIFIED		166.4	206.975	166.4	206.975	PASS				XYtable reached the target welding position
27/08/2026 15:39:47	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	3						3			Welding program was set on the welding machine
27/08/2026 15:39:47	20260827_153648_95690		1003195	PROGRAM_VERIFIED	3						3		3	Welding program was verified before actual welding
27/08/2026 15:39:47	20260827_153648_95690		1003195	WELD_STARTED	3	166.4	206.975							Start the welding process
27/08/2026 15:39:48	20260827_153648_95690		1003195	LIGHT_CURTAIN_ACTIVATED										Light curtain switched ON during execution
27/08/2026 15:39:49	20260827_153648_95690		1003195	LIGHT_CURTAIN_DEACTIVATED										Light curtain switched OFF
27/08/2026 15:39:54	20260827_153648_95690		1003195	RECOVERY_STARTED	3	166.4	206.975							Start the recovery process after power/safety interruption
27/08/2026 15:39:54	20260827_153648_95690		1003195	SUPERVISOR_AUTHORIZATION_REQUESTED	3	166.4	206.975							Waiting for supervisor to come and input a PASSCODE to unlock the machine
27/08/2026 15:39:58	20260827_153648_95690		1003195	SUPERVISOR_AUTHORIZATION_GRANTED	3	166.4	206.975							Machine unlocked for emergency recovery
27/08/2026 15:40:00	20260827_153648_95690		1003195	RESTART_POSITION_CONFIRMED	3	166.4	206.975							After successful confirmation, the machine will home and move to restart position
27/08/2026 15:40:02	20260827_153648_95690		1003195	RECOVERY_HOMING_STARTED	3	0	0							XYtable moves to home position after power/safety interruption
27/08/2026 15:40:13	20260827_153648_95690		1003195	RECOVERY_POSITION_VERIFIED	3	166.4	206.975	166.4	206.975	PASS				XYtable successfully reached the last target recovery position
27/08/2026 15:40:13	20260827_153648_95690		1003195	RECOVERY_COMPLETED	3	166.4	206.975							Welding process recovered to interrupted welding position
27/08/2026 15:40:13	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	3						3			Welding program was set on the welding machine
27/08/2026 15:40:13	20260827_153648_95690		1003195	PROGRAM_VERIFIED	3						3		3	Welding program was verified before actual welding
27/08/2026 15:40:13	20260827_153648_95690		1003195	WELD_STARTED	3	166.4	206.975							Start the welding process
27/08/2026 15:40:14	20260827_153648_95690		1003195	WELD_RESULT	3						3		3 OK	Welding completed successfully
27/08/2026 15:40:15	20260827_153648_95690		1003195	MOVE_STARTED		166.4	230.425							XYtable starts moving to the next welding position
27/08/2026 15:40:15	20260827_153648_95690		1003195	POSITION_VERIFIED		166.4	230.425	166.4	230.425	PASS				XYtable reached the target welding position
27/08/2026 15:40:16	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	4						4			Welding program was set on the welding machine
27/08/2026 15:40:16	20260827_153648_95690		1003195	PROGRAM_VERIFIED	4						4		4	Welding program was verified before actual welding
27/08/2026 15:40:16	20260827_153648_95690		1003195	WELD_STARTED	4	166.4	230.425							Start the welding process
27/08/2026 15:40:17	20260827_153648_95690		1003195	WELD_RESULT	4						4		4 OK	Welding completed successfully
27/08/2026 15:40:18	20260827_153648_95690		1003195	MOVE_STARTED		166.4	253.875							XYtable starts moving to the next welding position
27/08/2026 15:40:18	20260827_153648_95690		1003195	POSITION_VERIFIED		166.4	253.875	166.4	253.875	PASS				XYtable reached the target welding position
27/08/2026 15:40:18	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	5						5			Welding program was set on the welding machine
27/08/2026 15:40:19	20260827_153648_95690		1003195	PROGRAM_VERIFIED	5						5		5	Welding program was verified before actual welding
27/08/2026 15:40:19	20260827_153648_95690		1003195	WELD_STARTED	5	166.4	253.875							Start the welding process
27/08/2026 15:40:19	20260827_153648_95690		1003195	LIGHT_CURTAIN_ACTIVATED	5									Light curtain switched ON during execution
27/08/2026 15:40:20	20260827_153648_95690		1003195	LIGHT_CURTAIN_DEACTIVATED	5									Light curtain switched OFF
27/08/2026 15:40:25	20260827_153648_95690		1003195	RECOVERY_STARTED	5	166.4	253.875							Start the recovery process after power/safety interruption
27/08/2026 15:40:25	20260827_153648_95690		1003195	SUPERVISOR_AUTHORIZATION_REQUESTED	5	166.4	253.875							Waiting for supervisor to come and input a PASSCODE to unlock the machine
27/08/2026 15:40:28	20260827_153648_95690		1003195	SUPERVISOR_AUTHORIZATION_GRANTED	5	166.4	253.875							Machine unlocked for emergency recovery
27/08/2026 15:40:30	20260827_153648_95690		1003195	RESTART_POSITION_CONFIRMED	5	166.4	253.875							After successful confirmation, the machine will home and move to restart position
27/08/2026 15:40:35	20260827_153648_95690		1003195	RECOVERY_HOMING_STARTED	5	0	0							XYtable moves to home position after power/safety interruption
27/08/2026 15:40:49	20260827_153648_95690		1003195	RECOVERY_POSITION_VERIFIED	5	166.4	253.875	166.4	253.875	PASS				XYtable successfully reached the last target recovery position
27/08/2026 15:40:49	20260827_153648_95690		1003195	RECOVERY_COMPLETED	5	166.4	253.875							Welding process recovered to interrupted welding position
27/08/2026 15:40:49	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	5						5			Welding program was set on the welding machine
27/08/2026 15:40:49	20260827_153648_95690		1003195	PROGRAM_VERIFIED	5						5		5	Welding program was verified before actual welding
27/08/2026 15:40:49	20260827_153648_95690		1003195	WELD_STARTED	5	166.4	253.875							Start the welding process
27/08/2026 15:40:51	20260827_153648_95690		1003195	WELD_RESULT	5						5		5 OK	Welding completed successfully
27/08/2026 15:40:51	20260827_153648_95690		1003195	MOVE_STARTED		166.4	277.325							XYtable starts moving to the next welding position
27/08/2026 15:40:52	20260827_153648_95690		1003195	POSITION_VERIFIED		166.4	277.325	166.4	277.325	PASS				XYtable reached the target welding position
27/08/2026 15:40:52	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	6						6			Welding program was set on the welding machine
27/08/2026 15:40:52	20260827_153648_95690		1003195	PROGRAM_VERIFIED	6						6		6	Welding program was verified before actual welding
27/08/2026 15:40:52	20260827_153648_95690		1003195	WELD_STARTED	6	166.4	277.325							Start the welding process
27/08/2026 15:40:53	20260827_153648_95690		1003195	WELD_RESULT	6						6		6 OK	Welding completed successfully
27/08/2026 15:40:54	20260827_153648_95690		1003195	MOVE_STARTED		166.4	300.775							XYtable starts moving to the next welding position
27/08/2026 15:40:54	20260827_153648_95690		1003195	POSITION_VERIFIED		166.4	300.775	166.4	300.775	PASS				XYtable reached the target welding position
27/08/2026 15:40:55	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	7						7			Welding program was set on the welding machine
27/08/2026 15:40:55	20260827_153648_95690		1003195	PROGRAM_VERIFIED	7						7		7	Welding program was verified before actual welding
27/08/2026 15:40:56	20260827_153648_95690		1003195	WELD_STARTED	7	166.4	300.775							Start the welding process
27/08/2026 15:40:58	20260827_153648_95690		1003195	LIGHT_CURTAIN_ACTIVATED										Light curtain switched ON during execution
27/08/2026 15:41:01	20260827_153648_95690		1003195	LIGHT_CURTAIN_DEACTIVATED										Light curtain switched OFF
27/08/2026 15:41:01	20260827_153648_95690		1003195	RECOVERY_STARTED	7	166.4	300.775							Start the recovery process after power/safety interruption
27/08/2026 15:41:01	20260827_153648_95690		1003195	SUPERVISOR_AUTHORIZATION_REQUESTED	7	166.4	300.775							Waiting for supervisor to come and input a PASSCODE to unlock the machine
27/08/2026 15:41:04	20260827_153648_95690		1003195	SUPERVISOR_AUTHORIZATION_GRANTED	7	166.4	300.775							Machine unlocked for emergency recovery
27/08/2026 15:41:07	20260827_153648_95690		1003195	RESTART_POSITION_CONFIRMED	7	166.4	300.775							After successful confirmation, the machine will home and move to restart position
27/08/2026 15:41:09	20260827_153648_95690		1003195	RECOVERY_HOMING_STARTED	7	0	0							XYtable moves to home position after power/safety interruption
27/08/2026 15:41:22	20260827_153648_95690		1003195	RECOVERY_POSITION_VERIFIED	7	166.4	300.775	166.4	300.775	PASS				XYtable successfully reached the last target recovery position
27/08/2026 15:41:22	20260827_153648_95690		1003195	RECOVERY_COMPLETED	7	166.4	300.775							Welding process recovered to interrupted welding position
27/08/2026 15:41:22	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	7						7			Welding program was set on the welding machine
27/08/2026 15:41:22	20260827_153648_95690		1003195	PROGRAM_VERIFIED	7						7		7	Welding program was verified before actual welding
27/08/2026 15:41:22	20260827_153648_95690		1003195	WELD_STARTED	7	166.4	300.775							Start the welding process
27/08/2026 15:41:23	20260827_153648_95690		1003195	WELD_RESULT	7						7		7 OK	Welding completed successfully
27/08/2026 15:41:24	20260827_153648_95690		1003195	MOVE_STARTED		146.09	289.055							XYtable starts moving to the next welding position
27/08/2026 15:41:25	20260827_153648_95690		1003195	POSITION_VERIFIED		146.09	289.055	146.0875	289.05	PASS				XYtable reached the target welding position
27/08/2026 15:41:25	20260827_153648_95690		1003195	PROGRAM_SELECTION_STARTED	8						8			Welding program was set on the welding machine
27/08/2026 15:41:25	20260827_153648_95690		1003195	PROGRAM_VERIFIED	8						8		8	Welding program was verified before actual welding
27/08/2026 15:41:25	20260827_153648_95690		1003195	WELD_STARTED	8	146.09	289.055							Start the welding process
27/08/202														

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



- NOK welding results shall also be recorded.
- The same established data recording library / method as used for the manual welding process shall be used.

Assessment Result: The validation confirmed that the semi-automatic welding process records the Work Order, UDI / Serial Number and the related welding data using the same established data recording method as the manual welding process. Welding data are recorded for both OK and NOK welding results and remain traceable to the related battery pack.

E22

Not used = no content.

E23 – Final Product Validation

Objective: To verify that the modified semi-automatic resistance spot welding process consistently produces battery packs meeting all specified product quality requirements.

The validation confirms that the implemented process controls verified in E17 to E21 result in conforming products and that the transition from the manual welding process to the semi-automatic welding process does not negatively affect product quality.

Background: The previous evidence sections verify the individual process controls implemented in the modified welding machine, including workpiece preparation, welding position, welding parameters, error handling, process recovery and traceability.

This final validation verifies the quality of the finished battery packs produced by the modified manufacturing process.

Validation Procedure: Five battery packs shall be manufactured using the semi-automatic welding process under normal production conditions.

Each battery shall complete the fully automatic welding sequence consisting of **42 welding operations** performed on both sides of the battery pack (2x 21 welding's).

The finished battery packs shall then be evaluated according to the following validation activities.

Test	Expected Result (Evidence)	Result
Complete automatic welding cycle	Five battery packs successfully complete the full automatic welding sequence without process errors. Table E23-1	☑ PASS
Visual weld inspection	All welds comply with the defined visual acceptance criteria. Figure E23-1 and Inspection Report E23-1	☑ PASS
Welding monitor evaluation	All 42 welding operations of each battery pack are recorded as OK and meet the specified welding limits. Report E23-2	☑ PASS
Pull test verification	Pull testing is performed on all 5 packs x 42 welds = 210 pull tests. All measured values meet the specified acceptance criteria. Table E23-2	☑ PASS

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine

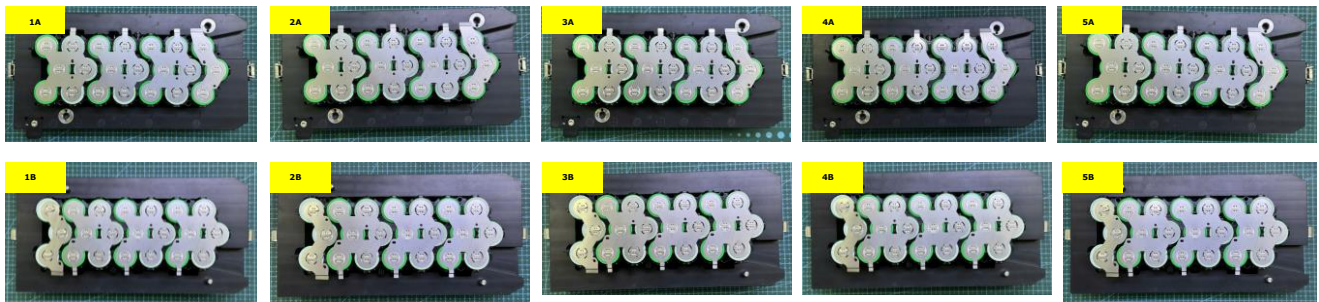


Metallographic section analysis	One representative weld of each different cell tab design is section cut and evaluated. The section confirms correct weld nugget formation / acceptable penetration and no damage to the cell can. Figure E23-2 and Report E23-3	☒ PASS
Final product evaluation	All five battery packs meet the specified product quality requirements and are accepted. Table E23-3	☒ PASS

Inspection Report E23-1 – Visual Weld Inspection

Battery No.	Weld Side	Weld Position	Acceptance Criteria	Result	Inspector	Remarks
1	Front / Rear	All	No visible defects, correct weld appearance	☒ PASS	Anh Vu	none
2	Front / Rear	All	No visible defects, correct weld appearance	☒ PASS	Anh Vu	none
3	Front / Rear	All	No visible defects, correct weld appearance	☒ PASS	Anh Vu	none
4	Front / Rear	All	No visible defects, correct weld appearance	☒ PASS	Anh Vu	None
5	Front / Rear	All	No visible defects, correct weld appearance	☒ PASS	Anh Vu	none

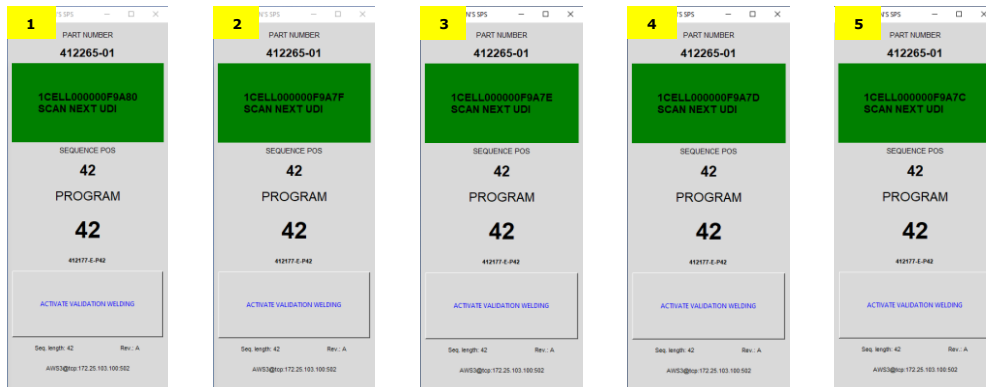
Below the photos of Battery 1 ... Battery 5 side A & B



Report E23-2 – Welding Monitor Evaluation

Battery No.	Total Welds	OK Welds	NOK Welds	Out of Limits	Parameter Violations	Report Attached below the table	Result
1	42	42	0	0	0	☒	☒ PASS
2	42	42	0	0	0	☒	☒ PASS
3	42	42	0	0	0	☒	☒ PASS
4	42	42	0	0	0	☒	☒ PASS
5	42	42	0	0	0	☒	☒ PASS

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



Above the snapshot of during the Soft-PLC receiving OK welding result (same like in the manual welding = already validated)

Table E23-2 – Pull Test Results (210 Welds) in Unit “N”

Battery No.	Welds Tested	Min. Pull Force	Max. Pull Force	Average	Failures YES / NO	Detailed Report	Result
1	42	41.409	212.314	133.754	No	Attached	☑ PASS
2	42	43.419	228.348	112.751	No	Attached	☑ PASS
3	42	57.913	215.637	136.775	No	Attached	☑ PASS
4	42	45.423	204.101	119.688	No	Attached	☑ PASS
5	42	41.182	213.514	134.227	No	Attached	☑ PASS

Report E23-3 – Metallographic Section Analysis

Cell Tab Type	Battery No.	Weld Position	Weld Nugget	Penetration	Cell Can Damage	Cracks / Voids	Result
Type A 211583-01-3L	1	1	☑ OK	☑ OK	☑ None	☑ None	☑ PASS
Type B 212724-02-6L	2	2	☑ OK	☑ OK	☑ None	☑ None	☑ PASS
Type C 212723-01-6R	3	22	☑ OK	☑ OK	☑ None	☑ None	☑ PASS
Type D 212323-01-3R	4	28	☑ OK	☑ OK	☑ None	☑ None	☑ PASS

Machine Safety & Validation: Semi-Automatic Resistance Spot Welding Machine



Section-cut photos as **Figure E23-2**.



Table E23-3 – Final Product Evaluation

Battery No.	Process Validation	Visual Inspection	Welding Monitor	Pull Test	Section Analysis	Final Evaluation
1	PASS	PASS	PASS	PASS	PASS	ACCEPTED
2	PASS	PASS	PASS	PASS	PASS	ACCEPTED
3	PASS	PASS	PASS	PASS	PASS	ACCEPTED
4	PASS	PASS	PASS	PASS	PASS	ACCEPTED
5	PASS	PASS	PASS	PASS	PASS	ACCEPTED

Acceptance Criteria

- Five consecutive battery packs shall successfully complete the automatic welding process.
- All **42 welding operations** of each battery pack shall satisfy the welding monitor acceptance criteria.
- All welds shall pass the defined visual inspection.
- All pull test results shall comply with the specified acceptance criteria.
- Metallographic section analysis shall confirm acceptable weld quality for each different cell tab design without evidence of excessive penetration, weld-through, or cell can damage.
- All five battery packs shall comply with the applicable drawing and product quality requirements.

Assessment Result: Five battery packs were manufactured using the modified semi-automatic welding process and evaluated by visual inspection, welding monitor analysis, pull testing and metallographic section analysis. All validation activities confirmed compliance with the specified product quality requirements. The results demonstrate that the modified welding process is capable of consistently producing conforming battery packs suitable for serial production.